Analytical Chemistry Cumulative Exam
September 6, 2024
"Something for Everyone"

Instructions: Answer the questions below. There is no extra credit for solving more than asked.

I. Warm-up. Pick 3 of the 6 in this section to solve. Be sure to write the number in your blue book.

1) Calculate the pH of a 1.6 M HNO_3.

2) The solubility of AgCl (K_{sp} = 1.8 x 10^{-10}) in a 0.03 M $NaNO_3$ solution will be (greater than, less than, equal to) the solubility of AgCl in a 0.01 M $Mg(ClO_4)_2$ solution. Justify your choice.

3) (True/ False) The pH measured using a standard glass electrode of 0.1M HCl in a 1.0 M NaCl solution equals the pH of 0.1 M HCl in a 0.01 M NaCl solution. Justify your choice.

4) Write the charge balance for an aqueous solution containing 0.1 M phthalic acid, a diacid, (abbreviate H_2Ph in your answer) in distilled, deionized water.

5) Estimate the pH expected for an aqueous solution containing 1.0 nM HNO_3. Justify your choice.

6) What is the difference between the equivalence point and the endpoint of a titration?

Intermediate: Pick 1of the 2 in this section and denote the number in your blue book.

7) Consider a potassium ion-selective electrode. This potassium electrode has a selectivity coefficient (k ($_{K+,Na+}$)) for Na^+ of 1 x 10^{-5} . If the Na^+ concentration of biological fluids is 150 mM and K^+ is 4 mM. Should you be concerned about using this electrode for measurement of K^+ in these solutions? Why or why not?

8) A) What is the typical expected pH of water from a distilled deionized water unit? Explain/Justify your answer.

B) Write the chemical equations describing the equilibrium chemistry in water in a beaker sitting out on a standard laboratory bench.

Problems

Number 9 must be attempted. I have now given this problem 3, possibly 4 times in the last year and no one has been able to solve this. This is sophomore analytical chemistry – maybe CHEM 2 Honors/Majors Chem.

9) You have water samples with unknown concentrations between 10 and 25 mM for each ion of OH-,

CO_3^{2-}, and HCO_3-.

For water, $K_w = 10^{-14}$ For H_2CO_3, pK_{a1} = 6.4 and pK_{a2} = 10.3.

Describe how you would quantify these three ions within this unknown sample. Assume you can easily access >100 mL of sample and have access to any wet chemistry techniques and glassware that you need, e.g., glassware, strong acids or bases, weak acids or bases, balances, and pH meters.

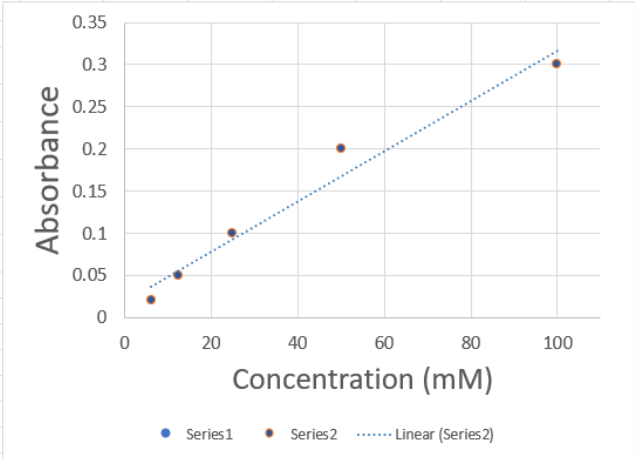
Very clearly denote what information you must obtain to determine the concentration of each of these in the unknown sample. If any spreadsheet process is needed, explain what must be coded or performed with the spreadsheet calculations.

Pick 1 from the following 2 problems to solve.

10) In your blue book, sketch a corresponding residual plot for the data below. Properly label your graph.

How can looking at both sets of data help you identify any potential problems with your calibration?

Concentration	Absorbance
6.125	0.02
12.5	0.05
25	0.1
50	0.2
100	0.3



11) A student was preparing solutions of theobromine (from chocolate) for an experiment. For theobromine, the molar mass is 180.16 g/mol.

A weigh by difference approach was used. This means a piece of weighing paper is tared to zero and then 180.41 mg of theobromine was obtained. This is then transferred to a volumetric flask and the paper is re-weighed to determine the residual leftover. After re-weighing the weigh paper, the remaining residual was 0.25 mg. The balance has a tolerance of 0.05 mg.

This solution was transferred to a 100.00 ± 0.08 mL volumetric flask.

a) What is the weight of theobromine added to the flask with appropriate error propagation?

b) What is the mM concentration of the solution with appropriate error propagation?